



2025

M24

BOOKLET NO.

105128

Forest Services

Animal Husbandry and Veterinary Science

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining questions, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note – Candidates will be allowed to use Scientific (Non-programmable type) calculators.

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SECTION – A

Q1. Write short notes on any five out of seven :	(8×5=40)
Q1. (a) Energy requirement for growth in dairy cattle.	8
Q1. (b) Evaluation of semen quality of animals.	8
Q1. (c) Maintenance of lactation.	8
Q1. (d) Management of animals under natural calamities.	8
Q1. (e) Methods of selection for breeding animals.	8
Q1. (f) General functions of minerals in animal body.	8
Q1. (g) Repeatability in animal breeding.	8
Q2. (a) Elucidate the role of vitamins in the animal body.	15
Q2. (b) State systems of breeding in animals and write detail about systems of inbreeding.	15
Q2. (c) Define ethology and elucidate environmental factors and regulatory mechanism involved in animal ethology.	10
Q3. (a) Discuss in detail the measures of food energy for animals.	15
Q3. (b) Write the differences between mitosis and meiosis in animal genetics.	15
Q3. (c) Write a note on development of practical and economic ration for pigs.	10
Q4. (a) Enlist the digestive organs of ruminants and describe their functions with respect to gastrointestinal secretions.	15
Q4. (b) Discuss different aspects of budgeting to provide the general framework for a dairy enterprise of 100 buffalo herd.	15
Q4. (c) Describe protein metabolism and synthesis in ruminants.	10

**SECTION – B**

Q5.	Solve any five out of seven :	(8×5=40)
Q5.	(a) Define lameness. Classification of lameness and enlist the forms of limb contact.	8
Q5.	(b) Discuss HCN poisoning in a cow with reference to etiology, sources, mechanism of toxicity, clinical signs, diagnostic approach, therapeutic management and control.	8
Q5.	(c) Enlist central and provincial Acts and explain prevention and control of infectious and contagious diseases in Animals Act 2009 (PCICD Act 2009).	8
Q5.	(d) Define abattoir. Discuss in detail anti-mortem examination of food animals along with judgement.	8
Q5.	(e) Discuss any four principles of programme planning.	8
Q5.	(f) Discuss about the origin and insertion of pudendal nerve and its significance in veterinary practice. (male and female)	8
Q5.	(g) Enlist different nerve blocks and write in detail about Peterson's nerve block in cattle.	8
Q6.	(a) Discuss the mechanism of respiration and exchanges of gases in detail.	15
Q6.	(b) Define pasteurization of milk. Discuss in detail HTST pasteurization of milk with its advantages and disadvantages over LTLT pasteurization.	15
Q6.	(c) Explain the role of animals and birds in prevalence and transmission of zoonotic diseases.	10
Q7.	(a) Describe the process of haemoglobin synthesis and its clinical significance.	15
Q7.	(b) Discuss in detail the etio-epidemiology, clinical signs, diagnosis, prevention and control of Listeriosis in sheep.	15
Q7.	(c) Discuss in detail rigor mortis and post-mortem pH decline during the conversion of muscles into meat.	10
Q8.	(a) Define shock. Classify the types of shock and describe in detail the pathophysiology, clinical signs, diagnosis and management of shock in animals.	15
Q8.	(b) Discuss commonly occurring production and deficiency diseases in buffalo. Describe etiopathogenesis, signs and symptoms, clinical findings, diagnosis and treatment of traumatic reticulo peritonitis in ruminants.	15
Q8.	(c) How extension teaching methods selected ? Discuss the uses, advantages and limitations of the individual contact methods.	10

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